Informatics

Binary search on sorted data

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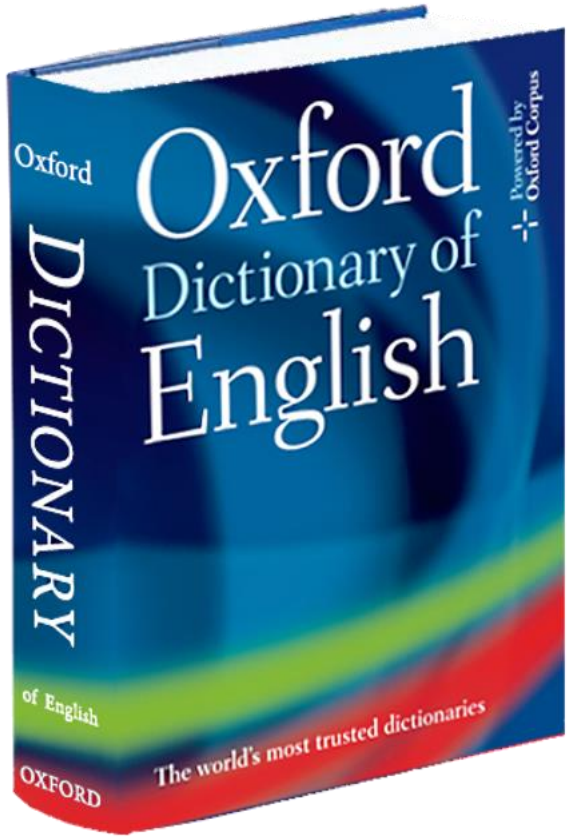
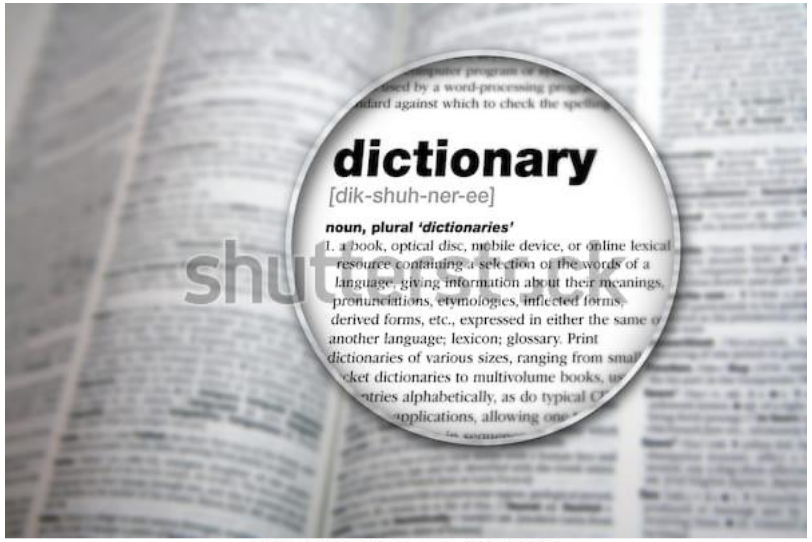
Looking for a word in a dictionary





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Looking for a word in a dictionary

Algorithm – first attempt

1. You are looking for the word *thing*
2. Open the dictionary at some position
3. If the word is in the open pages stop and *return the position*
4. If the word is *smaller* than the smaller word in the open pages
5. then open the dictionary at some position preceding the current one
6. else open the dictionary at some position following the current one
7. go back to step 3

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5. then open the dictionary at some position preceding the current one
6. else open the dictionary at some position following the current one
7. go back to step 3

What's wrong with this algorithm?

Looking for a word in a dictionary
Algorithm – second attempt

- You are looking for the word *thing*, *the pages to search are from the start to the end of the dictionary*
- Repeat while there are pages to search
  - Open the dictionary at some position inside the pages to search
  - If *thing* is in the open pages
    - stop and *return the position of the open pages*
  - If the word is *smaller* than the smaller word in the open pages
    - open the dictionary at some position preceding the current one: the pages to search are from the current start to the preceding position
  - else
    - open the dictionary at some position following the current one: the pages to search are from the following position to the current end
- stop and return not found

Looking for a word in a dictionary
Algorithm – second attempt

- You are looking for the word *thing*, *the pages to search are from the start to the end of the dictionary*
- Repeat while there are pages to search
  - Open the dictionary at some position inside the pages to search
  - If *thing* is in the open pages
    - stop and *return the position of the open pages*
  - If the word is *smaller* than the smaller word in the open pages
    - open the dictionary at some position preceding the current one
  - else
    - open the dictionary at some position following the current one
- stop and return not found

exit point: found

exit point: not found

How to test if *"there are pages to search"*

- Define the search area with two variables
  - start of the search area
  - end of the search area
  - after each failed search inside the open pages update either the start or the end of the search area

Looking for a word in a dictionary
Algorithm – third attempt

- You are looking for the word *thing*, *the pages to search are from the start to the end of the dictionary*
- Repeat while *the set of pages to search is not empty*
  - Open the dictionary at some position inside the pages to search
  - If the word is in the open pages
    - stop and *return the position of the open pages*
  - If *thing* is *smaller* than the smaller word in the open pages
    - update the end of the search area to the pages immediately before the open pages
  - else
    - update the start of the search area to the pages immediately after the open pages
- stop and return not found

Looking for a word in a dictionary
Algorithm – third attempt

- You are looking for the word *thing*, *the pages to search are from the start to the end of the dictionary*
- Repeat while *the set of pages to search is not empty*
  - Open the dictionary at some position inside the pages to search
  - If the word is in the open pages
    - stop and *return the position of the open pages*
  - If *thing* is *smaller* than the smaller word in the open pages
    - update the end of the search area to the pages immediately before the open pages
  - else
    - update the start of the search area to the pages immediately after the open pages
- stop and return not found

now we need only this "open"

left part	center	right part
start	*(start+end)/2*	*end*

how can we specify now the test

the set of pages to search is not empty?

Looking for a word in a dictionary
Algorithm – fourth attempt

- You are looking for the word *thing*,
- *set start to the first page of the dictionary*
- *set end to the last page of the dictionary*
- Repeat while *start <= end*
  - Open the dictionary at some position inside the pages to search
  - If the word is in the open pages
    - stop and *return the position of the open pages*
  - If *thing* is *smaller* than the smaller word in the open pages
    - update the end of the search area to the pages immediately before the open pages
  - else
    - update the beginning of the search area to the pages immediately after the open pages
- stop and return not found

Looking for a word in a dictionary
Algorithm – fourth attempt

- You are looking for the word *thing,*
- *set start to the first page of the dictionary*
- *set end to the last page of the dictionary*
- Repeat while *start* <= *end*
  - Open the dictionary at some position between start and end
  - If the word is in the open pages
    - stop and *return the position of the open pages*
  - If *thing* is *smaller* than the smaller word in the open pages
    - update the end of the search area to the pages immediately before the open pages
  - else
    - update the beginning of the search area to the pages immediately after the open pages
- stop and return not found

how can we specify now

at some position between start and end?

We could use any point between start and end, but the **best strategy** is using the **central point**!

In other words, the algorithm works with any choice, but using the center it **works better**

Challenge: **can you prove it?**

Looking for a word in a dictionary
Algorithm – final

- You are looking for the word *thing,*
- *set start to the first page of the dictionary*
- *set end to the last page of the dictionary*
- Repeat while *start* <= *end*
  - Open the dictionary in the position center = (start+end)/2
  - If the word is in the open pages
    - stop and *return the position of the open pages*
  - If *thing* is *smaller* than the smaller word in the open pages
    - update the end of the search area to the pages immediately before the open pages
  - else
    - update the beginning of the search area to the pages immediately after the open pages
- stop and return not found

Searching a target in a sorted (e.g. alphabetized) list: binary search

- Sequential search
  - it can be done on a list, no requirement on sorting
- If the list is sorted the search can be done in a more efficient way

```
parameters:

searchList: vector containing the
   sorted values to be searched

target: value to search

use:

start, end: integers indicating the
    start and the ending indexes of
    the scope

center: index of the center of the
    current scope

result:

found: boolean

position: index where value is equal to
    target, if found, or where the
    target could be inserted keeping the
    list sorted, if not found
```

```
- set start, end to the first and last element of searchList
- repeat if start<=end  (i.e. the scope is not empty)
  - compute center as (start+end)/2 using integer arithmetic
  - if target==searchList[center]
    - found: exit
    else
      if target < searchList[center]
        - continue search with scope start:center-1
      else
        - continue search with scope center+1:end
- if searchList[center] < target
  - the output position is center+1
  else the output position is center
- output: not found
```

initialization

```
- set start, end to the first and last element of searchList
- repeat if start<=end  (i.e. the scope is not empty)
  - compute center as (start+end)/2 using integer arithmetic
  - if target==searchList[center]
    - found: exit
    else
      if target < searchList[center]
        - continue search with scope start:center-1
      else
        - continue search with scope center+1:end
- if searchList[center] < target
  - the output position is center+1
  else the output position is center
- output: not found
```

main loop

finalization

Why should we use integer arithmetic?
What happens if start+end is odd?

left part	center	right part
start	*(start+end)/2*	*end*

Binary search (1)

Target: Eddy

	searchList	
1	Ann	start
2	Eddy	
3	Fred	
4	Mary	center
5	Oliver	
6	Peter	
7	Terry	
8	Victoria	end

```
algo:
- set start, end to the first and
  last element of searchList
- repeat if start<=end
  i.e. the scope is not empty
  - compute center as (start+end)/2
    using integer arithmetic
  - if target==searchList[center]
    - found: exit
    else
      if target < searchList[center]
        - continue search with scope
          start:center-1
      else
        - continue search with scope
          center+1:end
- if searchList[center] < target
  - the output position is center+1
  else the output position is center
- output: not found
```

Binary search (2)

Target: Eddy

	searchList	
1	Ann	start
2	Eddy	center
3	Fred	End
4	Mary	
5	Oliver	
6	Peter	
7	Terry	
8	Victoria	

```
algo:
- set start, end to the first and
  last element of searchList
- repeat if start<=end
  i.e. the scope is not empty
  - compute center as (start+end)/2
    using integer arithmetics
  - if target==searchList[center]
    - found: exit
    else
      if target < searchList[center]
        - continue search with scope
          start:center-1
      else
        - continue search with scope
          center+1:end
- if searchList[center] < target
  - the output position is center+1
  else the output position is center
- output: not found
```

Binary search (3)

Target: **Ada**

	searchList	
1	Ann	Start
2	Eddy	
3	Fred	
4	Mary	Center
5	Oliver	
6	Peter	
7	Terry	
8	Victoria	End

```
algo:
- set start, end to the first and
  last element of searchList
- repeat if start<=end
  i.e. the scope is not empty
  - compute center as (start+end)/2
    using integer arithmetics
  - if target==searchList[center]
    - found: exit
    else
      if target < searchList[center]
        - continue search with scope
          start:center-1
      else
        - continue search with scope
          center+1:end
- if searchList[center] < target
  - the output position is center+1
  else the output position is center
- output: not found
```

Binary search (4)

Target: Ada

	searchList	end
1	Ann	Start
2	Eddy	Center
3	Fred	End
4	Mary	
5	Oliver	
6	Peter	
7	Terry	
8	Victoria	

```
algo:
- set start, end to the first and
  last element of searchList
- repeat if start<=end
  i.e. the scope is not empty
  - compute center as (start+end)/2
    using integer arithmetics
  - if target==searchList[center]
    - found: exit
    else
      if target < searchList[center]
        - continue search with scope
          start:center-1
      else
        - continue search with scope
          center+1:end
- if searchList[center] < target
  - output position is center+1: exit
  else output position is center: exit
- target not found: exit
```

Binary search

Target: Ada

	searchList	
1	Ann	Start End Center
2	Eddy	
3	Fred	
4	Mary	
5	Oliver	
6	Peter	
7	Terry	
8	Victoria	

```
algo:
- set start, end to the first and
  last element of searchList
- repeat if start<=end
  i.e. the scope is not empty
  - compute center as (start+end)/2
    using integer arithmetics
  - if target==searchList[center]
    - found: exit
    else
      if target < searchList[center]
        - continue search with scope
          start:center-1
      else
        - continue search with scope
          center+1:end
- if searchList[center] < target
  - the output position is center+1
  else the output position is center
- output: not found
```

Binary search

Target: Ada

	searchList	End
1	Ann	Start
2	Eddy	
3	Fred	
4	Mary	
5	Oliver	
6	Peter	
7	Terry	
8	Victoria	

```
algo:
- set start, end to the first and
  last element of searchList
- repeat if start<=end
  i.e. the scope is not empty
  - compute center as (start+end)/2
    using integer arithmetics
  - if target==searchList[center]
    - found: exit
    else
      if target < searchList[center]
        - continue search with scope
          start:center-1
      else
        - continue search with scope
          center+1:end
- if searchList[center] < target
  - the output position is center+1
  else the output position is center
- output: not found
```

If the list has n elements and

- we repeat the execution several times with different targets,
- we keep track of the number of loop iterations in each experiment

<u>what is the maximum number of loop iterations we should expect?</u>

Search area size along the iterations

Search area size – n=1023	Iteration	log_2(n)-iteration
1023	1	9
511	2	8
255	3	7
127	4	6
63	5	5
31	6	4
15	7	3
7	8	2
3	9	1
1	10	0

In general, the number of iteration is

```
ceiling(log2(n+1))
```

Binary search is a

divide-and-conquer

algorithm

Recap:

What is the purpose of this algorithm?

- input t, v
- set f to 0
- repeat varying i from 1 to length of v
  - if the position i of v is equal to t
    - set f to i and stop repeating
- if f is zero
  - display "no""
- else
  - display the value of f

Recap:

If the length of v is **n**, what is the **maximum** number of repetitions?

- input t, v
- set f to 0
- repeat varying i from 1 to length of v
  - if the position i of v is equal to t
    - set f to i and stop repeating
- if f is zero
  - display "no""
- else
  - display the value of f

Recap:

If the length of v is **n**, what is the **minimum** number of repetitions?

- input t, v
- set f to 0
- repeat varying i from 1 to length of v
  - if the position i of v is equal to t
    - set f to i and stop repeating
- if f is zero
  - display "no""
- else
  - display the value of f

Recap:

If the length of v is **n**, what is the **average** number of repetitions?

- input t, v
- set f to 0
- repeat varying i from 1 to length of v
  - if the position i of v is equal to t
    - set f to i and stop repeating
- if f is zero
  - display "no""
- else
  - display the value of f

Useful R functions

algorithms

```
list.files(path = "<path>")
           pattern = "*.txt")
```

- produces a vector with the names of the files in the *path* directory that satisfy the *pattern*

```
rep.int(<integer number>,<number of repetitions>)
```

- produces a vector with a repeated integer, first argument is the integer, second is the repetition
- suitable for vector initialization

```
all(<logical vector>)
```

- produces TRUE if and only if **all** the elements of the logical vector are TRUE
- suitable for comparing two vectors